Audit.sql

```sql
-- Tutorial 12.2, test P4: purchase order approvals
-- Population: all purchase orders; expected: no rows
WITH ApprovalExceptions AS (
    SELECT 'Self-approved' AS Test, po.PONumber, po.OrderTotal
    FROM PurchaseOrder AS po
    WHERE po.CreatedByEmployeeID = po.ApprovedByEmployeeID
    UNION ALL
    SELECT 'Above approver limit', po.PONumber, po.OrderTotal
    FROM PurchaseOrder AS po
        INNER JOIN Employee AS e
            ON e.EmployeeID = po.ApprovedByEmployeeID
    WHERE po.OrderTotal > e.MaxApprovalAmount
    UNION ALL
    SELECT 'Dated after approver left', po.PONumber, po.OrderTotal
    FROM PurchaseOrder AS po
        INNER JOIN Employee AS e
            ON e.EmployeeID = po.ApprovedByEmployeeID
    WHERE e.TerminationDate IS NOT NULL
        AND po.OrderDate > e.TerminationDate
)
SELECT Test, COUNT(*) AS Orders, ROUND(SUM(OrderTotal), 2) AS OrderValue
FROM ApprovalExceptions
GROUP BY Test
ORDER BY Orders DESC;
```

| | Test | Orders | OrderValue |
|---|---|---|---|
| 1 | Above approver limit | 13 | 705779.39 |
| 2 | Self-approved | 9 | 651830.75 |
| 3 | Dated after approver left | 3 | 15419.28 |

Execution finished without errors.
Result: 3 rows returned
At line 258:
WITH ApprovalExceptions AS (

*Outlined: the three tests, with the number of purchase orders each flags and their value. Most of the fourteen orders fail two tests.*